

Project Report

iRevolution: A Data-Driven Exploration of Apple's iPhone Impact in India Using Tableau

1. INTRODUCTION

1.1 Project Overview

This project focuses on creating an interactive dashboard for product data analysis using Tableau and embedding it into a Flask web application. It visualizes sales, price, ratings, and popularity of products. Users can filter and explore the data easily through the web interface. The project demonstrates the integration of BI tools with web technologies. It emphasizes usability, accessibility, and effective data storytelling. It is designed for academic demonstration and local deployment

1.2 Purpose

- Analyze product sales and pricing
- Visualize ratings and popularity
- Reduce manual data effort
- Make dashboards interactive
- Provide web access via Flask
- Demonstrate data storytelling

2.IDEATION PHASE

2.1 Problem Statement

Manual analysis of product sales and performance data is time-consuming and error-prone. Traditional reports fail to provide interactive insights. Stakeholders require a system that can present trends visually and allow quick comparisons. The absence of a centralized visualization tool makes decision-making difficult. There is also a need for a simple web interface to access these insights. This project addresses these challenges using Tableau and Flask integration.

2.2Empathy Map Canvas

- Identify user needs
- Understand user frustrations
- Determine user expectations
- Highlight user behavior
- Visualize user journey
- Focus on usability

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2.3 Brainstorming

- Generate dashboard ideas
- Decide types of charts
- Determine filters and interactivity
- Plan stories and sequences
- Discuss layout design
- Identify potential issues

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. User opens the web page
2. Views dashboard overview
3. Applies filters for insights
4. Navigates through story slides
5. Analyzes product metrics
6. Interprets results for decisions

3.2 Solution Requirement

Functional Requirements:

- Display product sales data
- Apply interactive filters
- Embed dashboards in web page
- Render stories with multiple slides
- Allow responsive design
- Publish dashboards to Tableau Public

. Non-Functional Requirements:

- Fast dashboard loading
- Cross-browser compatibility
- Mobile responsiveness
- Minimal resource usage
- User-friendly interface
- Secure local deployment

3.3 Data Flow Diagram (Explanation)

- 1.Data collected from Excel/CSV
- 2.Processed in Tableau
- 3.Dashboard published to Tableau Public
- 4.Flask server loads HTML page
- 5.HTML page embeds Tableau visualization
- 6.User interacts via browse

3.4 Technology Stack

- Tableau Desktop / Tableau Public
- Python 3.x
- Flask framework
- HTML, CSS, Bootstrap
- JavaScript (for Tableau API)
- Localhost web server

4.PROJECT DESIGN

4.1 Problem Solution Fit

The project provides a solution by converting raw product data into visual insights. Interactive dashboards allow users to analyze sales, prices, ratings, and popularity efficiently. Flask enables access through a simple web interface without installing Tableau. The system reduces manual effort and improves clarity in data interpretation. It meets academic requirements for data visualization and web deployment. The solution is lightweight and easy to deploy locally.

4.2 Proposed Solution

The proposed solution integrates Tableau dashboards with a Flask-based web application. Data from Excel/CSV is visualized in Tableau, and dashboards are published to Tableau Public. Embed codes are used in HTML templates served by Flask. Users can interact with charts, apply filters, and explore stories directly in a browser. The system ensures accessibility and responsive design. It provides a complete visualization workflow from data to web deployment

4.3 Solution Architecture

Data Source: Excel / CSV file
Visualization: Tableau Dashboard & Story
Published Layer: Tableau Public
Backend: Flask (app.py)
Frontend: HTML + CSS + Bootstrap
User: Browser view & interaction

5.PROJECT PLANNING & SCHEDULING

Phase 1: Data collection and cleaning
Phase 2: Dashboard creation in Tableau
Phase 3: Story creation for insights
Phase 4: Publish to Tableau in Flask

Phase 5: Embeb dashboard in Flask
Phase 6: Documentation & Presentation
Estimated Duration: 2–3 weeks.

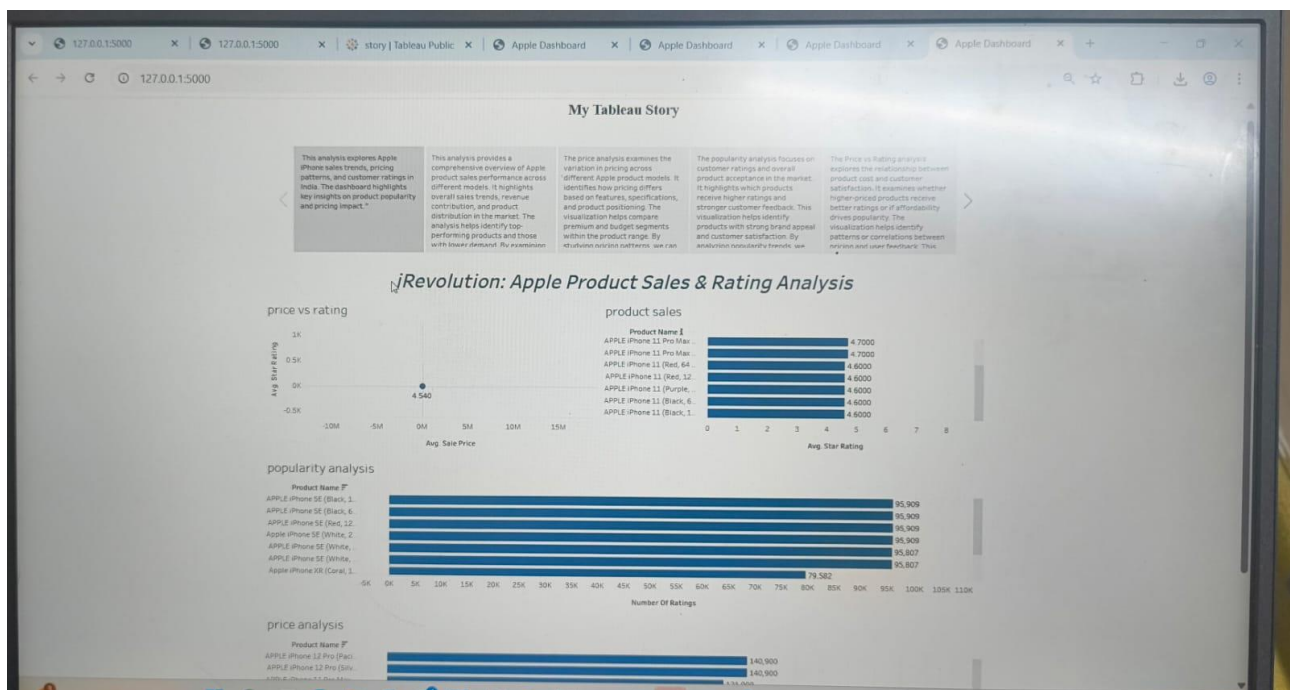
6.FUNCTIONAL AND PERFORMANCE TESTING

Performance Testing

- 1.Check dashboard loading speed
- 2.Test filter responsiveness
- 3.Monitor browser performance
- 4.Evaluate calculation fields
- 5.Test interactivity of story
- 6.Ensure no errors on local server.

7.RESULTS

7.1 Outputs:





8.ADVANTAGES & DISADVANTAGES

Advantages:

- Interactive visualizations
- Easy web access
- Reduces manual reporting
- Clear insights for decisions
- Lightweight deployment
- Academic demonstration ready

Disadvantages:

- Depends on internet for Tableau Public
- No authentication implemented
- Local deployment only

9.CONCLUSION

This project successfully demonstrates the creation of interactive dashboards and their integration with a web interface. It highlights the use of Tableau for data visualization and Flask for web deployment. The system allows users to analyze product performance easily and efficiently. It reduces manual reporting time and improves decision-making. The project provides a simple, functional solution suitable for academic purposes. Future improvements can include authentication and cloud deployment.

10.FUTURE SCOPE

- Add user login and authentication
- Connect to live database
- Deploy on cloud platforms
- Include more dashboards and analytics
- Enhance UI with advanced templates
- Real-time data integration

11.APPENDIX

Github link: <https://github.com/Sruthikanta94/iRevolution-A-Data-driven-Exploration-of-Apple-s-iPhone-Impact-in-India-using-Tableau>

